

"Falcon 2.0T: Cylinder 1 misfire from cracked ignition coil insulation"

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Area: bulletins

1 Condition

Falcon 2.0T vehicles may set diagnostic trouble code P0301 (Cylinder 1 Misfire Detected) with a flashing or steady malfunction indicator lamp and a noticeable rough idle. The misfire is typically most pronounced under load or in damp conditions. Confirm P0301 is cylinder-specific and not part of a multi-cylinder pattern before proceeding. Misfire isolation guidance is provided in the Falcon Engine Service Manual :: Misfire Diagnosis section.

2 Cause

The misfire originates from cracked insulation on the original ignition coil, part number IC-2042-A, which permits spark arc-over to the cylinder head under high secondary voltage demand. The coil has been superseded by the revised IC-2042-B, which uses an improved epoxy insulation jacket to resist cracking. Coil identification and supersession history is detailed in the Heron Ignition Service Manual :: Coil Identification and Supersession . Background on coil construction appears in the Heron Ignition Service Manual :: Ignition Coils chapter.

3 Repair Procedure

Replace the cylinder 1 ignition coil with the superseded IC-2042-B and install a new spark plug SP-1108, following procedure PROC-IGN-COIL for connector handling and torque values. Clear P0301 and verify misfire counters return to zero on a road test. The replacement steps are documented in the Heron Ignition Service Manual :: Ignition Coil Replacement Procedure , and spark plug gap specifications appear in the Heron Ignition Service Manual :: Spark Plug Selection and Specifications .

4 Parts Information

The superseding ignition coil IC-2042-B fully replaces the discontinued IC-2042-A; do not reuse or reorder the IC-2042-A part number. Each repair requires one IC-2042-B coil and one SP-1108 spark plug for the affected cylinder. Refer to the Heron Ignition Service Manual :: Spark Plugs section for plug application notes and the Falcon Engine Service Manual for engine-specific torque references.